

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Public Bus Live Tracking & Crowding Estimator

SRN	PES1UG24AM396
Project	LiveTrack (Jira key: LIV)
Source Requirements	Lab 1 — FR-001 to FR-005

1. Backlog: Epics and User Stories

The Lab 1 functional requirements were grouped into three Epics and broken down into eight User Stories, each estimated using Fibonacci story points and prioritized by user/business impact.

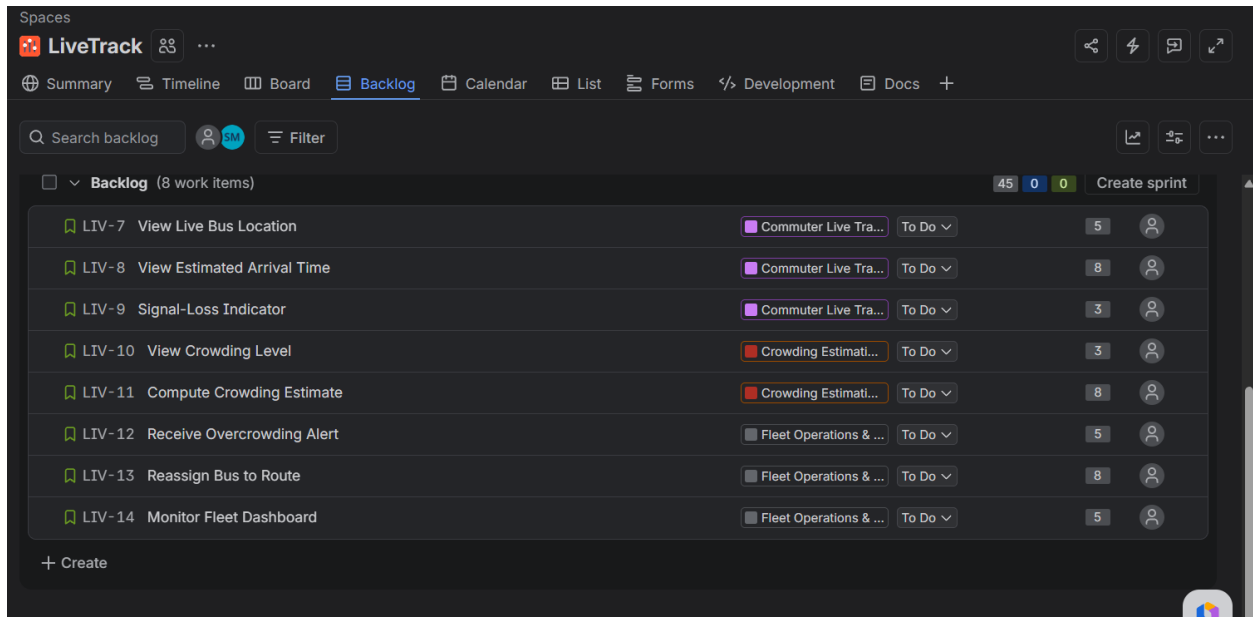


Figure 1 — Backlog view showing all 3 Epics and 8 User Stories with story points (total: 45 points).

ID	Story	Epic	Priority	Story Points
LIV-7	View Live Bus Location	Commuter Live Tracking & ETA	High	5
LIV-8	View Estimated Arrival Time	Commuter Live Tracking & ETA	High	8
LIV-9	Signal-Loss Indicator	Commuter Live Tracking & ETA	Medium	3
LIV-10	View Crowding Level	Crowding Estimation & Visibility	Medium	3
LIV-11	Compute Crowding Estimate	Crowding Estimation & Visibility	Medium	8
LIV-12	Receive Overcrowding Alert	Fleet Operations & Overcrowding Response	High	5
LIV-13	Reassign Bus to Route	Fleet Operations & Overcrowding Response	Medium	8
LIV-14	Monitor Fleet Dashboard	Fleet Operations & Overcrowding Response	Medium	5

2. Sprint 1 Execution

Sprint 1 (1-week duration) contained the commuter-facing core stories: View Live Bus Location, View Estimated Arrival Time, View Crowding Level, and Compute Crowding Estimate (24 story points).

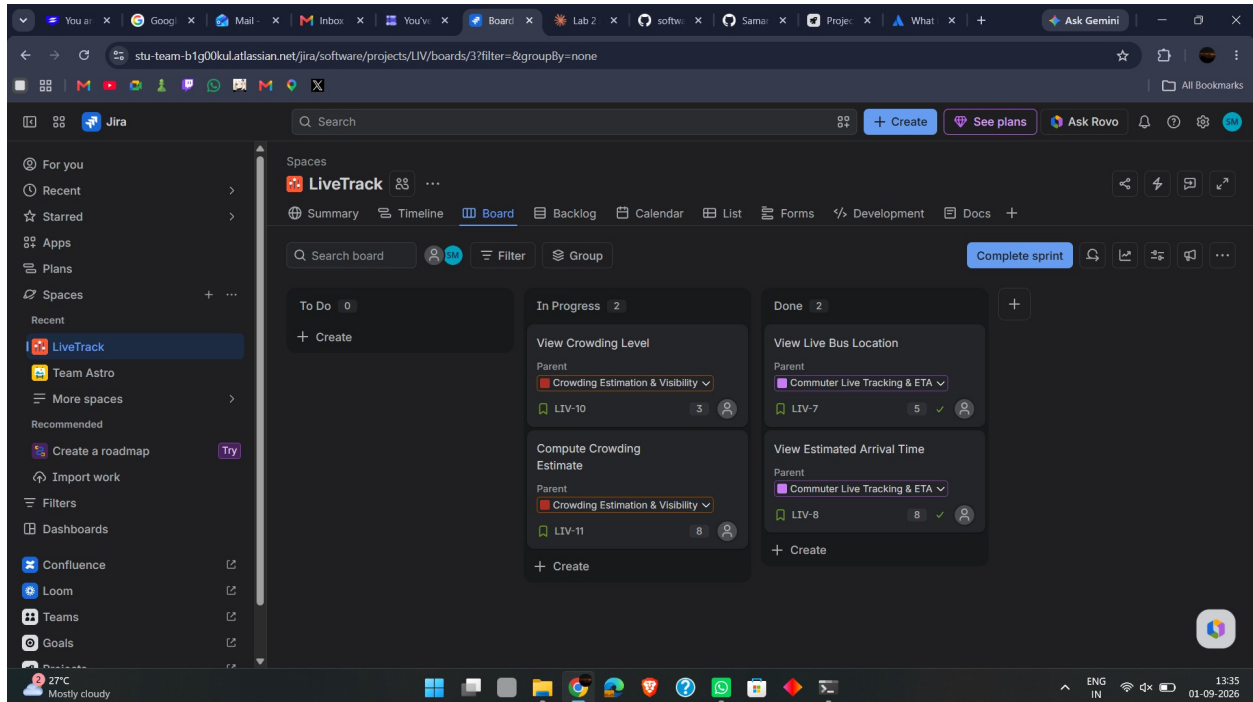


Figure 2 — Sprint 1 board mid-simulation: cards moved across To Do, In Progress, and Done as work was simulated.

All four Sprint 1 stories were moved through To Do → In Progress → Done and the sprint was completed.

3. Sprint 2 Execution

Sprint 2 (1-week duration) contained the fleet-controller and resilience stories: Signal-Loss Indicator, Receive Overcrowding Alert, Reassign Bus to Route, and Monitor Fleet Dashboard (21 story points).

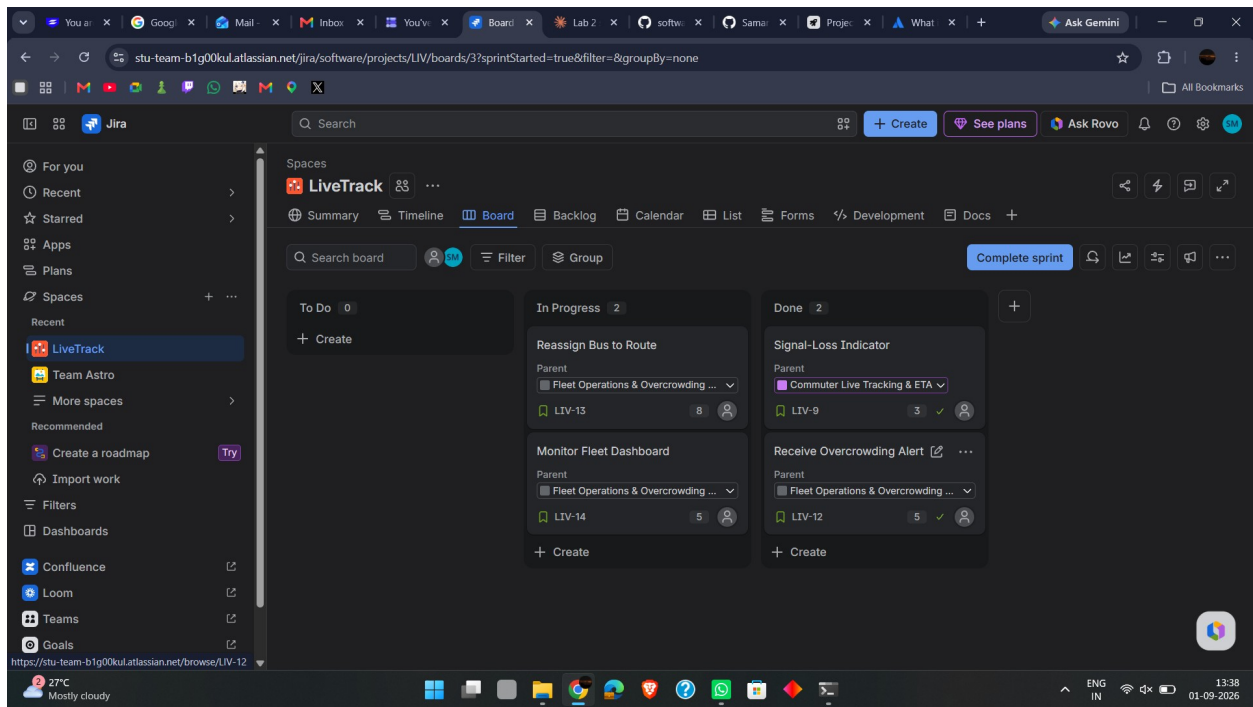


Figure 3 — Sprint 2 board mid-simulation: cards moved across To Do, In Progress, and Done as work was simulated.

All four Sprint 2 stories were moved through To Do → In Progress → Done and the sprint was completed.

4. Burndown Chart

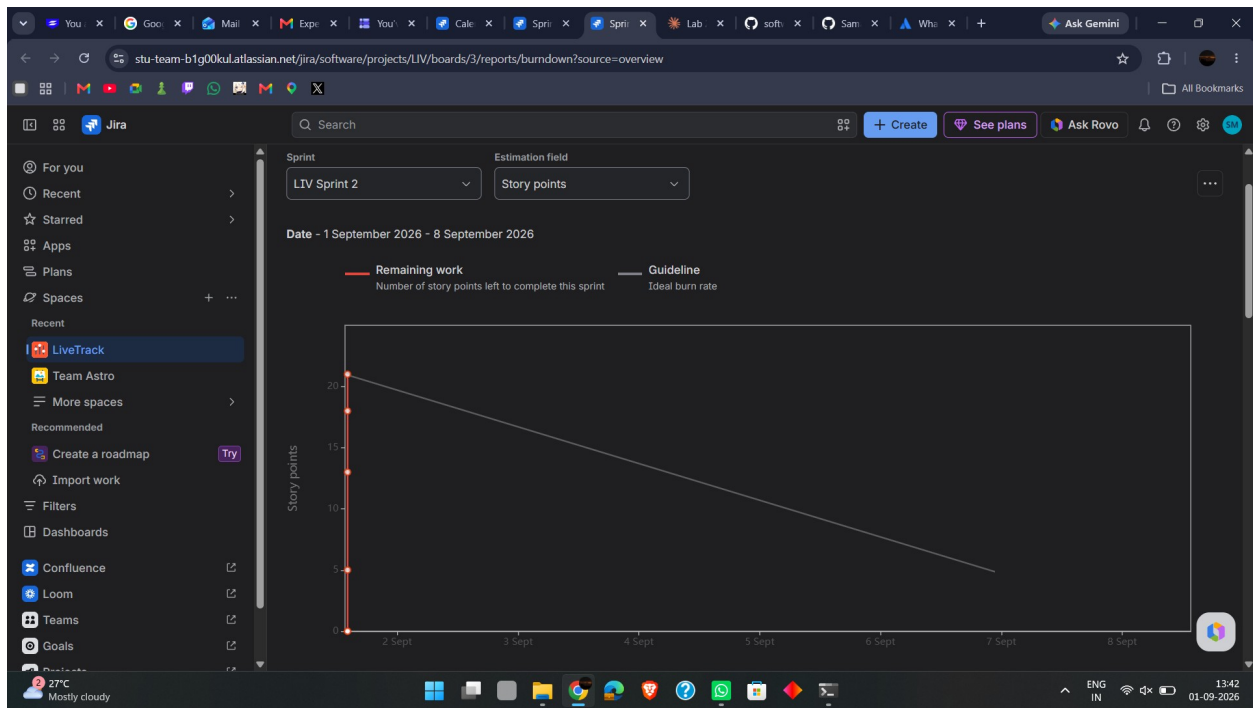


Figure 4 — Sprint burndown chart showing remaining story points against the ideal guideline.

5. Reflection Questions

1. Did your estimations reflect the actual effort?

The Fibonacci points assigned during planning (3, 5, and 8) were meant to capture relative complexity rather than literal hours, and in this simulation every story was completed regardless of its point value, so the estimates could not be validated against real effort. In a live team setting, the 8-point stories (View Estimated Arrival Time, Compute Crowding Estimate, Reassign Bus to Route) genuinely involve more moving parts — live GPS/traffic integration, sensor-data calculation, and cross-route re-scheduling — so the relative sizing is reasonable even though this simulation did not test it under time pressure.

2. Was your backlog well-prioritized?

Yes. The three High-priority stories (View Live Bus Location, View Estimated Arrival Time, Receive Overcrowding Alert) map directly to the two Lab 1 requirements marked High priority (FR-001, FR-002, FR-004), which represent the core commuter value proposition and the operational safety net for the fleet controller. The Medium-priority stories (crowding visibility, bus reassignment, dashboard, signal-loss handling) are all valuable but depend on the High-priority foundation being in place first, which is reflected in the Sprint 1/Sprint 2 split.

3. How did your simulated sprint align with your plan?

Both sprints completed all planned stories with no scope changes and no carry-over into the next sprint. Because the sprint was simulated rather than run over a real week, all eight stories were moved to Done within minutes of starting each sprint rather than being spread across the full sprint window.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart shows the remaining-work line dropping to zero almost immediately at the start of the sprint window instead of tracking the diagonal guideline line down gradually across the week. This is expected for a compressed simulation rather than a real sprint, but it highlights what the chart is designed to catch in a real team: if points are burned down far faster than the guideline, it is usually a sign the team under-committed during sprint planning relative to their real capacity — in a live project this would prompt pulling additional backlog items into the sprint rather than leaving capacity idle.